

CARMANIA— All Your Dream Cars at a Single Place

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Course

Bachelor of Technology (B.Tech) – Cyber Security

Subject

Frontend Fullstack Development

Institution

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ABSTRACT

Carmania is a web-based car customization platform developed using React. The application enables users to explore different car models and customize them based on various parameters such as color, wheels, and interior features. The system dynamically updates pricing based on user selections, providing a real-time and interactive experience.

The platform also allows users to save customized builds and manage them through a dedicated dashboard. Additionally, users can contact dealers with their selected configurations, making the application practical and user-friendly. This project demonstrates the implementation of modern frontend technologies to build a responsive and dynamic web application.

INTRODUCTION

With the rapid growth of digital platforms, users expect interactive and personalized experiences when exploring products online. In the automobile industry, most platforms provide only static information about vehicles without offering customization features.

Carmania addresses this limitation by providing a dynamic platform where users can explore cars and customize them based on their preferences. The application focuses on enhancing user interaction through real-time updates and an intuitive interface. It simulates real-world car configuration systems and provides a seamless experience for users.

PROBLEM DESCRIPTION

Traditional car browsing platforms lack interactivity and do not provide users with the ability to customize vehicles before making a decision. Users often face difficulty in understanding how different configurations affect pricing and features.

There is a need for a system that allows users to:

- Explore multiple car options efficiently
- Customize vehicles based on personal preferences
- View real-time price updates
- Save and manage customized builds

Carmania solves these problems by providing a complete and interactive car customization platform.

OBJECTIVES

- To develop an interactive platform for browsing car models
- To implement a dynamic car customization system
- To provide real-time price updates based on user selections
- To enable saving and managing customized builds

- To design a responsive and user-friendly interface
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TOOLS AND TECHNOLOGIES USED

The Carmania application was developed using the following technologies:

- **Frontend:** React (Vite)
- **Programming Language:** JavaScript (ES6+)
- **Routing:** React Router
- **State Management:** React Hooks
- **Storage:** LocalStorage
- **Styling:** CSS
- **Version Control:** Git & GitHub
- **Deployment:** Vercel

These tools enabled the development of a responsive and interactive web application.

SYSTEM ARCHITECTURE

The system follows a client-side architecture where all functionalities are handled within the frontend.

- React components manage UI and logic
- React Router handles navigation between pages
- LocalStorage stores user data such as saved builds
- The application is deployed on Vercel for public access

The architecture ensures smooth performance and scalability for future enhancements.

IMPLEMENTATION

The Carmania application consists of multiple components and pages:

- **Home Page:** Displays available car models
- **Dashboard (Garage):** Shows saved builds and statistics
- **Build Page:** Allows users to customize cars (color, wheels, interior)
- **Car Details Panel:** Displays specifications and features

The customization logic dynamically calculates pricing based on user-selected options. The application uses LocalStorage to persist data across sessions, ensuring a seamless user experience.

RESULT

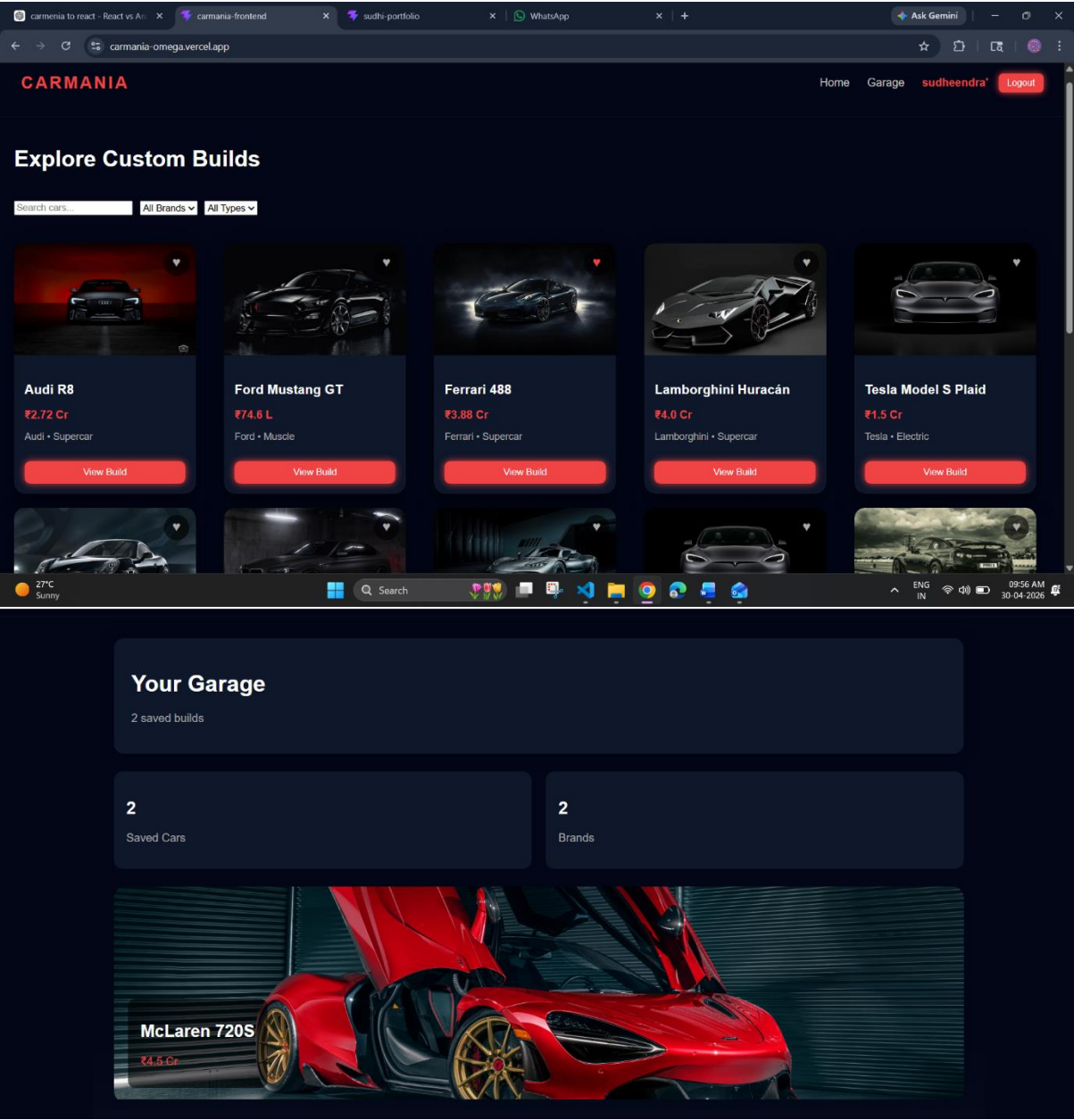
The Carmania application was successfully developed and deployed. The system allows users to:

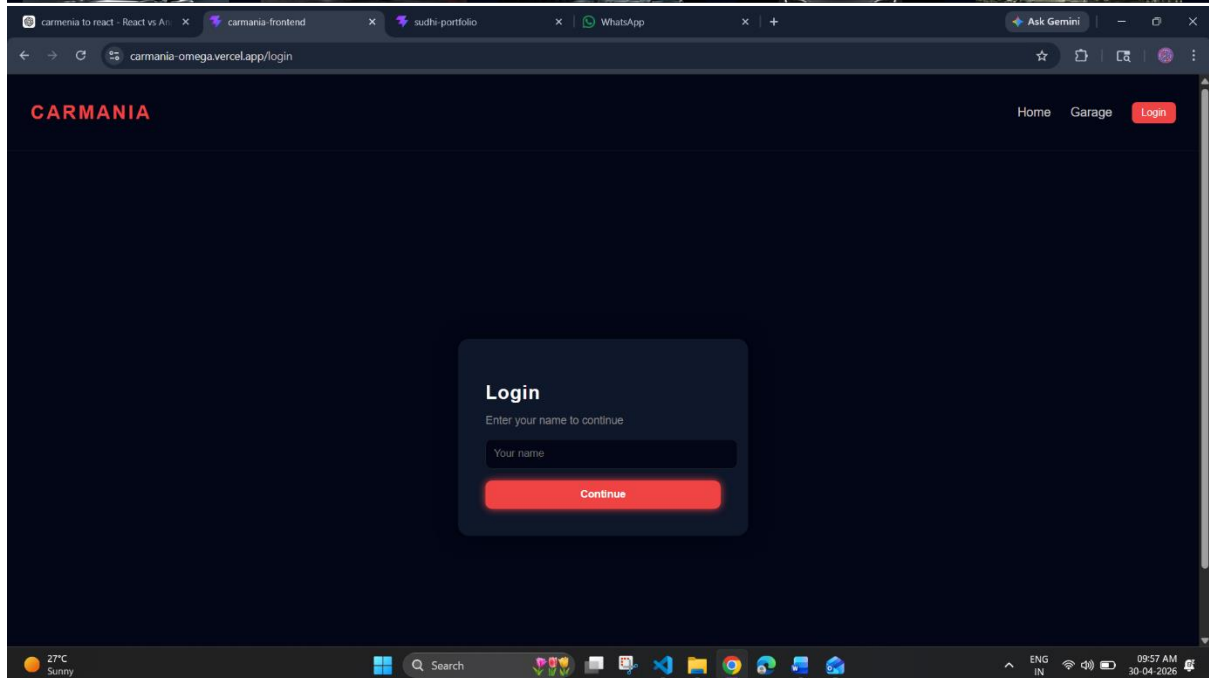
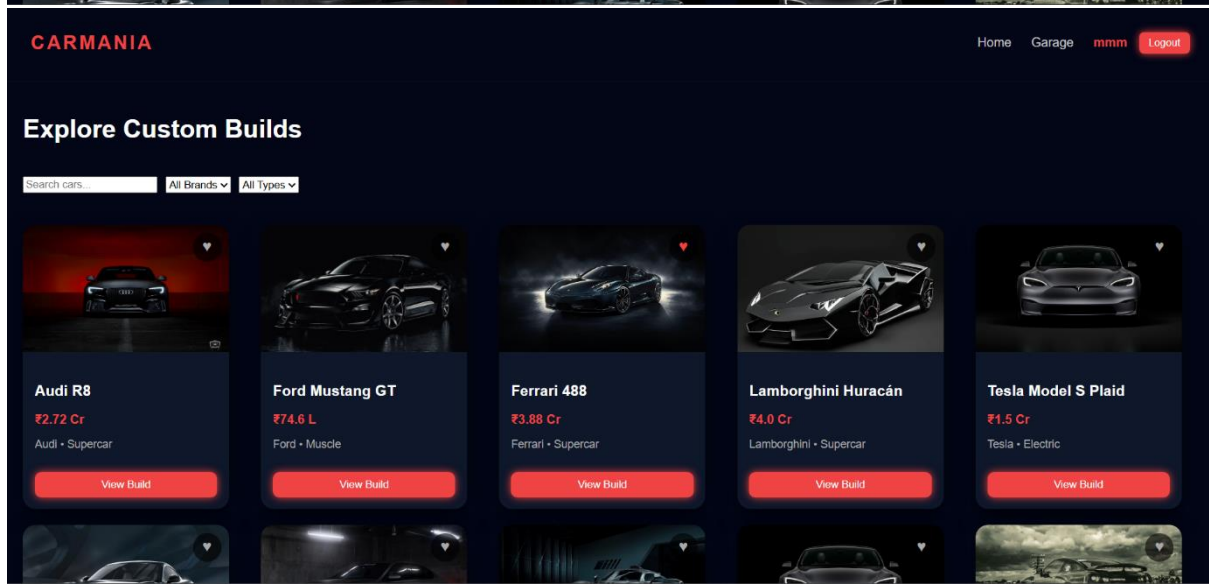
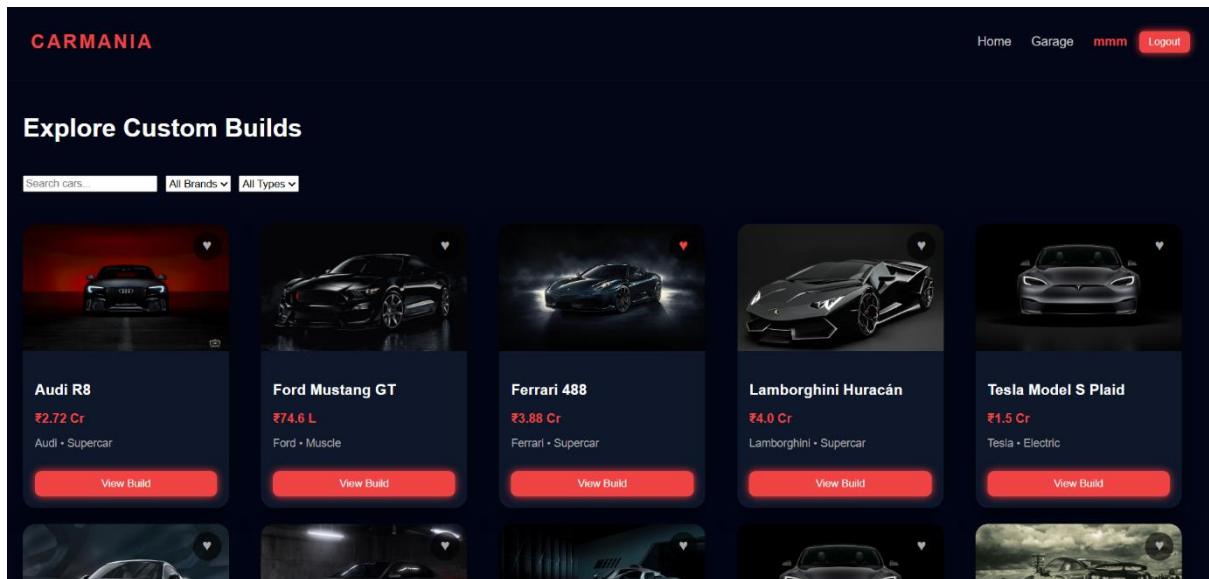
- Browse car models
- Customize vehicles
- View real-time price updates
- Save and manage builds
- Contact dealers with configuration details

Live Application:

<https://carmania-omega.vercel.app/>

Screenshots:





CONCLUSION

The Carmania project successfully demonstrates the development of a dynamic and interactive web application for car customization. The system enhances user experience by providing real-time feedback and intuitive controls.

The project highlights the importance of modern frontend technologies in building scalable and user-centric applications. It also serves as a strong foundation for future improvements and real-world implementations.

FUTURE SCOPE

The project can be further improved with:

- Backend integration (Node.js or Firebase)
- User authentication and account system
- Cloud database for storing user builds
- Advanced animations and UI enhancements
- AI-based car recommendations
- Payment and booking system

These enhancements can transform the platform into a full-scale commercial application.
